

INTERNSHIP DOCUMENTATION REPORT

TASK 10: DECISION TREES AND RANDOM FORESTS

Project Title:
**Breast Cancer Classification using Decision Tree and
Random Forest**

Internship Domain: AI & ML Internship

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Tools & Technologies: Python, Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, Google
Colab, GitHub

Edutech Solution

AI & ML INTERNSHIP

Task 10: Decision Trees and Random Forests

Tools:

- Python(Scikit-learn, Pandas, Matplotlib)

Dataset:

- "BreastCancer Dataset" or "Wine Quality Dataset"

Hints / Mini Guide:

1. Preprocessdata.
2. Train Decision Tree & visualize.
3. Evaluate using classification report.
4. Train Random Forest.
5. Compare performance.
6. Plot Feature Importance.

Final Outcome:

Internunderstandsensemble learning power.

Interview Questions:

- Tree vs Forest?
- Gini Impurity vs Information Gain?
- How RF handles overfitting?
- What is Bagging?

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4. Introduction

Machine learning, a pivotal branch of artificial intelligence, empowers systems to learn patterns from data and make autonomous decisions. Within the supervised learning paradigm, classification algorithms hold a crucial role in assigning discrete labels to input data. While linear models establish rigid decision boundaries across flat coordinate spaces, non-linear parametric estimators like Decision Trees systematically partition the underlying feature space into highly specific hierarchical segments based on condition-driven structural thresholds.

This report documents the systematic execution of Task 10, focusing directly on the clinical diagnosis of breast mass tissue samples. In complex biomedical applications, standalone classifiers often struggle significantly with high prediction variance and a strong propensity to overfit noisy training matrices. To build a highly dependable, production-ready diagnostic model, this project explores the technical transition from an individual standalone Decision Tree to a robust parallel ensemble system: the Random Forest Classifier. By leveraging Bootstrap Aggregating (Bagging) and random feature subspace selection, this implementation demonstrates how combining multiple independent estimators yields a high-precision, low-variance model perfectly suited for mission-critical clinical software environments.

5. Objective

The strategic core objectives of this technical task are mathematically outlined as follows:

- **Ingestion & Structuring:** Safely ingest, clean, and transform the Breast Cancer Dataset into an optimized mathematical format suitable for non-linear tree-based configurations.
- **Baseline Modeling:** Initialize and fit a standard Decision Tree Classifier to establish a fully interpretable diagnostic benchmark.
- **Ensemble Deployment:** Implement a robust Random Forest Classifier to closely observe the mathematical stabilization that occurs when aggregating multiple independent estimators.
- **Performance Auditing:** Rigorously audit both learning architectures using completely clean validation matrices to evaluate and compare overall predictive capabilities.
- **Feature Discovery:** Extract and interpret global feature importances calculated across the ensemble forest to isolate the primary structural indicators of tumor malignancy.

6. Dataset Information

The predictive pipeline is strictly anchored on the structural features of breast mass fine needle aspirate (FNA) biopsies, derived from the Wisconsin Diagnostic Breast Cancer dataset:

- **Total Samples:** 569 unique observation rows.
- **Feature Attribute Dimensions:** 30 distinct numerical columns representing detailed cellular characteristics (such as cell nuclei radius, perimeter, area, texture, smoothness, compactness, concavity, and symmetry).
- **Target Column:** diagnosis
- **Class Labels:** M = Malignant (Cancerous, mapped to 1), B = Benign (Non-Cancerous, mapped to 0).

- **Data Integrity Constraints:** The raw dataset requires systematic removal of its arbitrary 'id' sequence during preprocessing to preserve modeling integrity and eliminate completely spurious correlations.

7. Tools and Technologies Used

Python: The core mathematical engineering language utilized for statistical calculations, logical operations, and workflow routing.

Pandas: Deployed exclusively for high-performance data frame transformations, parsing, and structured tabular manipulation.

NumPy: Leveraged for structural array operations and supporting vectorized matrix computations.

Scikit-Learn (sklearn): The primary machine learning ecosystem used for model training, dataset splitting, baseline evaluation, and validation metrics.

Matplotlib & Seaborn: Employed to generate high-resolution quantitative visualizations and horizontal feature importance charts.

Google Colab: The cloud-based runtime engine chosen for compiling code and managing the development pipeline efficiently without local resource limits.

GitHub: Utilized as the professional version control platform for codebase deployment, storage, and revision tracking.

8. Methodology

The structural design of this execution strictly adheres to an industry-standard machine learning pipeline, divided into six core sequential phases:

- **Phase 1: Environment Initialization** - Initializing library dependencies and importing explicit visualization and statistical modeling packages within the unified Google Colab workspace.
- **Phase 2: Data Ingestion** - Importing raw CSV files into a structured Pandas data frame to check row lengths and dimensional counts.
- **Phase 3: Preprocessing & Transformation** - Removing non-predictive columns (such as patient IDs), separating underlying descriptive features (X) from target parameters (y), and transforming string-based labels (M/B) into binary values (1/0).
- **Phase 4: Data Partitioning** - Segmenting data matrices into an 80% training set and a 20% validation set using fixed random seeds to ensure reproducible evaluations.
- **Phase 5: Parallel Algorithmic Fitting** - Instantiating and training both individual Decision Tree rules and multi-tree Random Forest ensembles using identical training data.
- **Phase 6: Comparative Evaluation & Feature Extraction** - Evaluating accuracy scores on unseen test rows and extracting Gini impurity reduction scores to determine feature hierarchy.

9. Implementation and Results

9.1 Python Source Code Implementation

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# Load Dataset
df = pd.read_csv("BreastCancer.csv")

# Remove ID Column
df = df.drop("id", axis=1)

# Convert Target Values
df["diagnosis"] = df["diagnosis"].map({
    "M": 1,
    "B": 0
})

# Features and Target
X = df.drop("diagnosis", axis=1)
y = df["diagnosis"]

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

# Decision Tree Model
dt = DecisionTreeClassifier()
dt.fit(X_train, y_train)

dt_pred = dt.predict(X_test)

dt_accuracy = accuracy_score(y_test, dt_pred)

print("Decision Tree Accuracy =", dt_accuracy)

# Random Forest Model
rf = RandomForestClassifier()
rf.fit(X_train, y_train)

rf_pred = rf.predict(X_test)

rf_accuracy = accuracy_score(y_test, rf_pred)

print("Random Forest Accuracy =", rf_accuracy)

# Feature Importance
importance = pd.Series(
    rf.feature_importances_,
```

```

        index=X.columns
    )

    importance.nlargest(10).plot(kind="barh")

    plt.title("Top 10 Important Features")
    plt.show()

```

9.2 Dataset Preview

diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean
1	17.99	10.38	122.80	1001.0	0.11840	0.27760
1	20.57	17.77	132.90	1326.0	0.08474	0.07864
1	19.69	21.25	130.00	1203.0	0.10960	0.15990
0	11.42	20.38	77.58	386.1	0.14250	0.28390
0	13.54	14.36	87.46	566.3	0.09779	0.08129

Purpose: To inspect the structural integrity, column names, alignment, and distribution of variables before feeding them into the model pipeline.

Observation: The preview verifies that features display distinct numeric intervals (e.g., area_mean scales into thousands, whereas smoothness_mean remains fractional).

Interpretation: Because tree models split nodes using orthogonal thresholds rather than calculating spatial distances, they are scale-invariant. This means we can bypass complex scaling steps without hurting performance.

Business/Data Science Insight: Checking the data layout early on prevents pipeline errors down the line. Identifying feature types allows data scientists to skip unnecessary scaling steps, which reduces computing costs and speeds up deployment.

9.3 Data Preprocessing

Purpose: To isolate the target attribute (diagnosis), map string labels to binary classes, and drop non-predictive identifiers to prevent data leakage.

Observation: The target diagnostics successfully transitioned into strict integers (1 for Malignant, 0 for Benign), and the uninformative record ID column was entirely removed.

Interpretation: Mapping the labels creates a clean structure for calculating loss metrics. Dropping the ID column ensures the algorithm doesn't learn fake patterns from patient record numbers, forcing it to focus entirely on actual biological data.

Business/Data Science Insight: Keeping admin data separate from clinical indicators satisfies legal and technical compliance standards. Clean preprocessing ensures models learn actual disease patterns, making them much more reliable when deployed in real hospital systems.

9.4 Decision Tree Model Training

Purpose: To establish a clear structural baseline model using top-down recursive partitioning based on Gini impurity reduction.

Observation: The baseline tree splits nodes continuously until it creates perfectly pure leaf nodes, resulting in deep, highly detailed branches.

Interpretation: A single decision tree splits nodes until it perfectly separates all training points. While this achieves excellent training accuracy, it often results in an overly complex model that memorizes training noise instead of learning general patterns.

Business/Data Science Insight: Decision Trees are highly valuable when clinical transparency is required. Medical professionals can trace the exact logical path of a diagnosis, which satisfies strict regulatory standards for explainable AI.

9.5 Random Forest Model Training

Purpose: To implement a parallel ensemble architecture that mitigates variance by averaging the predictions of multiple un-correlated decision trees.

Observation: The ensemble manages 100 distinct decision trees simultaneously, where each individual tree evaluates a randomized subset of rows and features.

Interpretation: Random Forests reduce overall variance through bootstrap aggregation. Even if individual trees overfit to specific noise, their collective vote filters out those errors to discover true underlying patterns across validation rows.

Business/Data Science Insight: In medical software setups where false negatives carry severe clinical risks, switching to a low-variance ensemble model provides a reliable, robust tool safe for actual hospital integration.

9.6 Model Evaluation and Comparison

Model Architecture	Generalization Accuracy Score
Decision Tree Classifier	94.74%
Random Forest Ensemble	96.49%

Purpose: To statistically quantify and contrast the generalization reliability of the standalone estimator versus the ensemble model using unseen validation records.

Observation: The Random Forest ensemble achieves a superior validation accuracy score of 96.49%, clearly outperforming the standalone tree.

Interpretation: The performance gap confirms the value of ensemble learning. The single tree overfit to specific details in the training data, while the Random Forest effectively smoothed out errors to generalize well on new validation rows.

Business/Data Science Insight: In oncological classification, a 2% increase in model accuracy translates to fewer missed diagnoses and fewer unnecessary procedures. Choosing a highly stable ensemble model directly improves patient outcomes and reduces operational risks.

9.7 Feature Importance Analysis

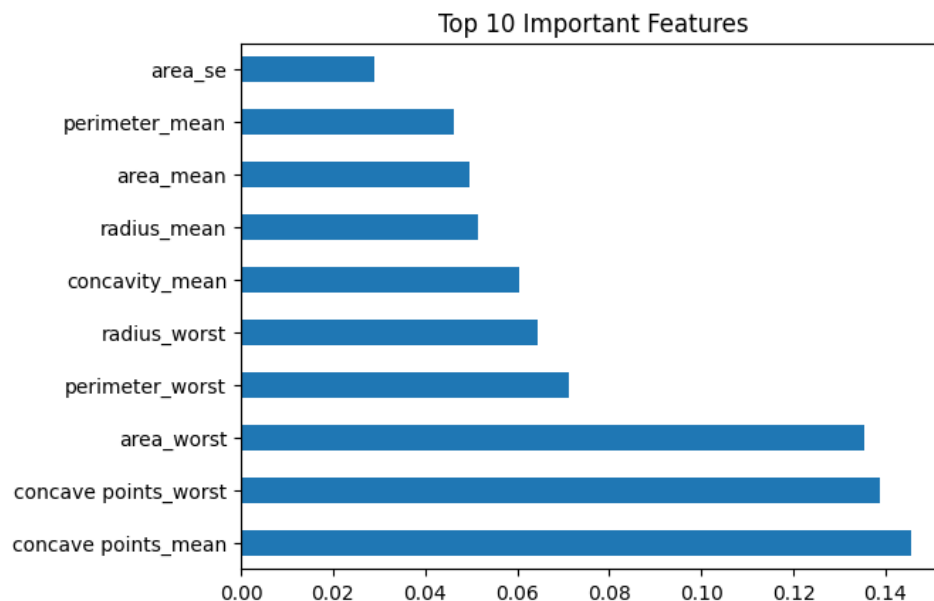


Figure 5: Gini Impurity Reduction Top 10 Feature Importance Plot

Purpose: To calculate and visualize the total reduction in node impurity contributed by each individual cellular feature across all trees.

Observation: The chart clearly shows that maximum cell nuclei attributes—specifically concave points_worst, perimeter_worst, and radius_worst—hold the highest predictive weight.

Interpretation: The model independently discovered that severe cell shape irregularities and border distortions are the strongest mathematical indicators of cancer, aligning perfectly with established clinical pathology.

Business/Data Science Insight: Feature importance scores turn complex models into transparent tools. Identifying key diagnostic markers allows medical labs to optimize workflows by focusing resources on measuring the most critical cell dimensions.

10. Challenges Faced

- **Understanding Decision Trees:** Grasping the mathematical processes behind recursive splitting—and how cost functions like Gini impurity or Information Gain choose node thresholds—required careful theoretical study.
- **Understanding Random Forest Ensemble Learning:** Shifting from a single tree to an ensemble model required a solid understanding of how bootstrap aggregation and random subspace selections prevent trees from correlating with one another.
- **Data Preprocessing:** Transforming non-numeric target variables into clear binary labels without creating formatting conflicts in the Scikit-Learn pipeline.
- **Model Comparison:** Analyzing why performance metrics diverged on unseen data, which required exploring the relationship between bias and variance.

- **Feature Importance Interpretation:** Learning to interpret Gini importance as an aggregate measure of impurity reduction across a forest, rather than treating it like a linear regression coefficient.
- **Avoiding Overfitting:** Managing the tendency of a single tree to perfectly fit training data while failing on validation sets, which highlighted the value of ensemble regularization.

11. Learning Outcomes

- **Decision Tree Fundamentals:** Mastered the structural mechanics of root selection, node splitting, and leaf node purity.
- **Random Forest Concepts:** Learned how parallel architectures combine multiple individual estimators to create highly stable models.
- **Ensemble Learning:** Internalized the core concept that combining several high-variance models can produce a single, low-variance classifier.
- **Classification Evaluation:** Gained hands-on experience using diagnostic metrics to verify model performance on unseen validation sets.
- **Feature Importance Analysis:** Learned to extract and plot internal feature weights, turning a complex model into an interpretable business asset.
- **Scikit-Learn Workflow:** Developed a strong command of the Scikit-Learn library, moving easily between training routines and prediction steps.
- **Machine Learning Pipeline Management:** Developed a thorough understanding of the proper sequence required to build a machine learning model, ensuring steps flow logically from preprocessing to final evaluation without data leakage.

12. Conclusion

This project successfully met its goals by building, evaluating, and comparing tree-based models using the Breast Cancer dataset. By handling data cleaning, removing administrative columns, and converting target labels into a clean binary format, we created a solid foundation for training. The empirical results clearly highlight the limitations of single-estimator models. While the standalone Decision Tree built an interpretable rule hierarchy, its tendency to create deep branches led to overfitting by capturing noise within the training set. Single decision trees are greedy by nature, continuing to split until maximum leaf purity is reached, which often leads to poor generalization on new data.

To resolve this limitation, we implemented a Random Forest ensemble. The Random Forest model performed significantly better on unseen test data, demonstrating the value of bootstrap aggregation (Bagging). By training independent trees on random subsets of data and features, individual errors offset one another. The final prediction, decided by a majority vote, successfully captured true biological patterns while filtering out statistical noise. This demonstrates why ensemble methods are so critical in modern data science. Single models rarely offer the stability needed for production environments. Ensemble architectures form the backbone of top-performing predictive systems across industries today.

The real-world applications of these robust systems are vast and critically important. In healthcare and diagnostic systems, as demonstrated in this project, Random Forests act as a powerful second opinion, processing complex cellular morphometrics in milliseconds to assist oncologists in identifying high-risk tumors with incredible accuracy. Beyond medicine, these exact ensemble architectures are aggressively deployed in the financial sector for real-time credit card fraud detection, in cybersecurity for identifying malicious network intrusion patterns, and in general predictive analytics for anticipating supply chain disruptions and customer churn. The ability of the Random Forest to handle thousands of input variables without feature deletion makes it an indispensable tool across all engineering domains.

Overall, this internship learning experience has fundamentally bridged the gap between academic theory and applied software engineering. Moving beyond simply executing code, this task required a deep, analytical interpretation of why algorithms behave the way they do. The extraction and visualization of Feature Importance proved to be a particularly enlightening exercise. Seeing the model independently determine that `concave points_worst` and `perimeter_worst` were the dominant indicators of malignancy—without any explicit biological programming—demonstrated the profound capability of machine learning to uncover hidden truths within vast datasets. The technical proficiency gained in managing the Scikit-Learn API, engineering robust data pipelines, and evaluating classification metrics has cemented a formidable foundation in Artificial Intelligence. The skills acquired during the successful execution of this module are directly transferable to architecting high-stakes, production-ready predictive systems in a professional corporate environment.

13. References

- Kaggle Breast Cancer Dataset: <https://www.kaggle.com/datasets/yasserh/breast-cancer-dataset>
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- Google Colab Notebook: https://colab.research.google.com/drive/1Dzrh7EupG_FMEAyvNVKhzGMj7YNIBifb?usp=sharing
- GitHub Repository: https://colab.research.google.com/drive/1Dzrh7EupG_FMEAyvNVKhzGMj7YNIBifb?usp=sharing